# *Full Stack Development with MERN*

# *Project Documentation format*

**(Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking**

# *PROJECT DOCUMENTATION*

## Full Stack Development (MERN)

## ❖ Project Title

**Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking**


**Team Leader:**

- Dharani Priya A– Frontend Developer

**Team Members:**

- Bhuvaneshwari S– Backend Developer
- Bhuvaneswari K – Database Manager
- Ananthi N– API Integration
- Kaavitha M– Testing & Documentation


## ❖ Abstract


❖ This project focuses on automating the process of importing large datasets into a system and establishing relationships between different data entities using techniques similar to Import Sets and Dot Walking.

❖ The system is built using the MERN stack (MongoDB, Express.js, React.js, Node.js), enabling efficient data ingestion, transformation, and relational mapping within a NoSQL environment.

## ❖ Objectives

❖ Automate bulk data import (CSV/Excel/JSON)
❖ Transform raw data into structured format
❖ Establish relationships between collections
❖ Reduce manual data handling
❖ Provide UI for monitoring imports and errors

## ❖ Concept Mapping (ServiceNow → MERN Equivalent)

| ❖ ServiceNow Concept | ❖ MERN Equivalent |
|---|---|
| ❖ **Import Set** | ❖ Staging Collection |
| ❖ **Transform Map** | ❖ Mapping Logic (Node.js) |
| ❖ **Coalesce** | ❖ Unique Key Matching |
| ❖ **Dot Walking** | ❖ Mongoose Populate |
| ❖ **Scheduled Import** | ❖ Cron Job |

## ❖ Database Design

❖ **Example Collections:**

❖ **Users Collection**

❖ {
"_id": "ObjectId",
"name": "John",
"email": "john@example.com"
}

❖ **Orders Collection**

❖ {
"_id": "ObjectId",
"userId": "ObjectId",
"product": "Laptop",
"price": 1000
}

❖ 👉 Relationship:
❖ Orders.userId → Users._id
❖ GET /api/errors

- **Relationship Mapping (Dot Walking Equivalent)**
- Order.find()
  .populate('userId') // dot walking equivalent
  .then(data => console.log(data));

## Testing

- **Unit Testing:**
- API endpoints
- Data transformation logic

- **Integration Testing:**
- Import → Transform → Store flow

- **Tools:**
- Jest / Mocha
- Postman
-

## Results

- Reduced manual data entry by ~80%
- Improved data accuracy
- Faster data processing time
-

## Challenges

- Handling large datasets
- Data validation complexity
- Maintaining relationships in NoSQL

## Future Enhancements

- Real-time data streaming
- AI-based data validation
- UI dashboards for analytics
- Multi-format import support

## ❖ References


- ❖ MERN Stack Documentation
- ❖ MongoDB Official Docs
- ❖ Node.js Documentation


## ❖ Appendix (Optional)

- ❖ Screenshots
- ❖ API responses
- ❖ Sample datasets